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CSCI2160F11Quiz8WritingMethods

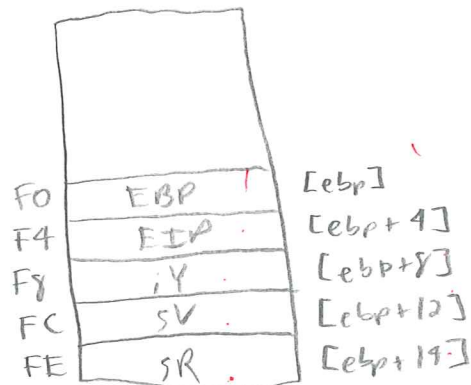
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There are problems on both sides of this quiz.

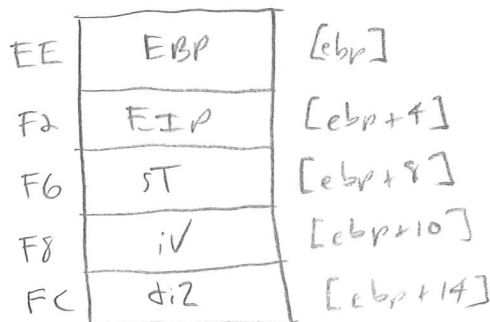
Draw the stack and label BOTH "sides" of your stack indicating the value of ESP and also references to the "names" of the variables within the stack using **EBP**. You may assume that the passed variables are "correct" in terms of what is expected.

iX dword 27
iY dword 35
iZ dword 18, 32, 96
sV word 22
sT word -1
sR word 7

1a) (10 pts) ; iX = fun1(iY, sV, sR)



b) (10 points) ; sR = fun2(sT, iV, &iZ) where & means "address of"



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2) (30 points) Write the method which would correspond to the UML statement below (Be sure to preserve all registers that require it).

sumup(x:int, y:int):int

it accepts two integers and returns their sum

proc sumup Near32 - 1

push ebp ✓

mov ebp, esp

~~sub esp, 8~~

xInt equ [ebp+12]

yInt equ [ebp+8]

mov eax, xInt

add eax, yInt

add esp, 8

pop ebp

ret

sumup endp

There are no local variables

; make room for local variables

; assign a name to first parameter

; assign a name to second parameter

; add up

; the two numbers

; clean up stack

why?

-3
[signature]

3) (50 points) Write the method which would return the total of all the values in an array of **ints** whose address is the first parameter, and whose second parameter is the number in the array. Be sure to preserve all registers that require it.

totalarray(&array, Num:int):int

proc totalarray Near32

push ebp

mov ebp, esp

push ebx

push ecx

xor eax, eax

mov ecx, [ebp+8]

mov ebx, [ebp+12]

sumup:

add eax, [ebx]

add ebx, 4

loop sumup

pop ecx

pop ebx

pop ebp

ret

totalarray endp

; preserve

; registers

; clear eax

; use array size as a counter

; assign starting position in array

; add element

; move array position

; loop until there are no more elements

; recover

; registers

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